Subrat Prasad Panda

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Research Statement: I am broadly interested in **Trustworthy Reinforcement Learning**, particularly in designing interpretable, robust, and verifiable policies for safety-critical domains. My current work focuses on **Neurosymbolic** approaches such as programmatic policies, combining explicit knowledge with learning to bridge System 1 (neural) and System 2 (symbolic) reasoning.

Education

2023 – . . . **Ph.D., Computer Science,** NTU Singapore
Thesis: *Programmatic Policies for Trustworthy Reinforcement Learning*
Advisors: *Arvind Easwaran, Blaise Genest*

2019 – 2021 **M.Tech., Computer Science,** ISI Kolkata, India
Thesis: *Strategies for User Allocation and Service Placement in Multi-Access Edge Computing*
CPI: 82.1% Advisor: *Ansuman Banerjee*

2012 – 2016 **B.Tech., Electronics and Instrumentation,** NIT Silchar, India
Thesis: *Real-Time Position Control of DC Motor using PID and Fuzzy Logic*
CPI: 7.92/10 Advisor: *Munmun Khanra*

Employment History

2022 – 2023 **Research Associate**, CNRS@CREATE, Singapore
- Designed a granular control system for microgrids using Deep Reinforcement Learning.

2021 – 2022 **Machine Learning Engineer**, InFocusp Innovation Pvt. Ltd., Pune, India
- Developed solutions for the Abstraction and Reasoning Challenge (ARC) using Reinforcement Learning.

2021 – 2021 **Research Engineer**, TiHAN, IIT Hyderabad
- Developed braking and throttle control systems using Arduino for a test vehicle.
- Deployed a lane detection model on the NVIDIA AGX board.

2016 – 2018 **Aviation Officer**, HPCL, Chandigarh International Airport
- Managed maintenance and repair of equipment for aircraft refueling operations.

2015 – 2015 **Summer Intern**, DRDO, New Delhi, India
- Enhanced colored image contrast in the spatial domain using histogram equalization and cubic spline interpolation.

Research Publications

Conference Proceedings

1. **S. P. Panda**, B. Genest, and A. Easwaran, "Reasoning with incremental knowledge for sample efficient hierarchical reinforcement learning," Under submission to AAMAS-26, 2026.
2. **S. P. Panda**, B. Genest, A. Easwaran, and P. N. Suganthan, "Vanilla gradient descent for oblique decision trees," in *ECAI 2024*, IOS Press, Oct. 2024. DOI: 10.3233/faia240607.
3. **S. P. Panda**, A. Banerjee, and A. Bhattacharya, "User allocation in mobile edge computing: A deep reinforcement learning approach," in *2021 IEEE International Conference on Web Services (ICWS)*, IEEE, 2021, pp. 447–458.
4. **S. P. Panda**, K. Ray, and A. Banerjee, "Service allocation/placement in multi-access edge computing with workload fluctuations," in *International Conference on Service-Oriented Computing*, Springer, 2021, pp. 747–755.

5 **S. P. Panda**, K. Ray, and A. Banerjee, "Dynamic edge user allocation with user specified qos preferences," in *International Conference on Service-Oriented Computing*, Springer, 2020, pp. 187–197.

Journal Articles

1 **S. P. Panda**, B. Genest, and A. Easwaran, "Approximation-free oblique decision trees," *Under submission to JMLR*, 2025.

2 **S. P. Panda**, B. Genest, A. Easwaran, R. Rigo-Mariani, and P. Lin, "Methods for mitigating uncertainty in real-time operations of a connected microgrid," *Sustainable Energy, Grids and Networks*, vol. 38, p. 101 334, 2024.

Workshops

1 **S. P. Panda**, B. Genest, and A. Easwaran, *Programmatic reinforcement learning for trustworthy microgrid management*, AI4X 2025 International Conference, 2025.

Interests and Skills

Research	Reinforcement Learning, Neuro-Symbolic AI, Knowledge and Reasoning
Programming	Python, PyTorch, MATLAB, LaTeX, ROS, …
Courses	Mathematical Statistics, Deep Learning, Machine Learning, Computer Vision

Miscellaneous Experience

Awards and Achievements

2023	**DesCartes Scholarship** — Ph.D. scholarship awarded by CNRS@CREATE under the DesCartes project.
2021	**M.Tech Best Thesis Nomination** — Ranked among the top 5 theses of the batch.
2019	**M.Tech Scholarship** — Scholarship awarded to pursue a master's degree.
2016	**Top 25 Finalist**, General Electric Edition Challenge — IoT-based engine exhaust gas monitoring system.
2015	1^{st} **Round**, Texas Instruments Innovation Challenge — Non-invasive digital blood pressure meter.

Mentoring and Supervision

F1/10th	**F1/tenth Racing Team, NTU** — Advised students on control and navigation strategies.
FYP	**Undergraduate Final Year Projects, NTU** — Guided projects on CNN robustness and programmatic policies using LLMs.
URECA	**Undergraduate Research, NTU** — Mentored projects on Decision Transformers, Mechanistic Interpretability, ARC Reasoning, and Physics-Inspired RL.

References

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